

PAUL ABHINAI DERISEMU

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SUMMARY

Accomplished data professional with over 5 years of robust experience as an ML Data Analyst at Amazon, complemented by a recent Master's degree in Data Analytics. Possesses a comprehensive skill set encompassing Data Analysis, Engineering, and Machine Learning. Proficient in Python, R, SQL, and advanced data visualization techniques, dedicated to deriving actionable insights. Demonstrated success in driving strategic decisions and pioneering innovative solutions in dynamic operational environments. Certified in TensorFlow, AngularJS, and Azure Data Science, poised to lead impactful data-driven initiatives and provide strategic leadership within progressive organizations.

- Proficient in SQL for querying, manipulating, and managing data, with over 5 years of hands-on experience.
- Strong understanding of relational database concepts and extensive use of Microsoft SQL Server in professional settings.
- Developed and optimized procedures for on-premises and real-time applications, showcasing leadership in project execution and team management.
- Solid grasp of fundamental Database Management System (DBMS) principles, effectively mentoring and supporting team members.
- Basic proficiency with NoSQL databases, particularly MongoDB, enhancing versatility in data management solutions and team collaboration.
- Familiar with AWS serverless services like Lambda and S3, contributing to scalable and efficient data operations and team productivity.
- Led initiatives in Extract, Transform, Load (ETL) processes and data integration using SQL Server Integration Services (SSIS), optimizing team workflows and performance.
- Experienced in creating insightful reports using SQL Server Reporting Services (SSRS), facilitating informed decision-making and team alignment.
- Demonstrated expertise in database design principles and normalization techniques, providing strategic guidance, and fostering a collaborative team environment.
- Led and participated in academic or small-scale projects focusing on database management, emphasizing teamwork, and achieving project goals.
- Committed to continuous professional development in advanced database technologies and business intelligence tools, enhancing team capabilities and productivity.
- Basic exposure to SQL Server Analysis Services (SSAS) for Online Analytical Processing (OLAP) and data mining, leveraging analytical insights to support team objectives.

TECHNICAL SKILLS

Programming Languages: Python, R, SQL, HTML, SAS(Basic)

Tools: Excel, Tableau, Power BI, SQL Server Management Studio, ArcGIS, Sklearn, Stats Model, Keras, Pandas, AWS, Pytorch, Tensorflow

Databases: MySQL, PostgreSQL (Basic)

Cloud Technologies: AWS EC2, AWS Lambda, S3, IAM

Machine Learning & Applications: Data Preprocessing, Supervised ,Unsupervised Learning, Deep Learning

Operating Systems: Mac, Windows, Linux

Certifications:Attained "Trustworthy AI for Decision-Making Systems" certification from George Washington University.

RELEVANT WORK EXPERIENCE

Amazon

Hyderabad, India

Sr ML-Data Analyst

June 2017-July 2022

- Contributed to data modeling initiatives by curating and provisioning user utterance datasets for Alexa, enabling the optimization and deployment of robust natural language processing models
- Employed the **ATAT tool** extensively to iteratively develop data models using user utterances. Integrated these models with Redshift servers to optimize functionality and improve operational performance."
- Was part of a development team that engineered performance tracking tools, specifically the **Goal Success Rate (GSR)**, which employs predictive analytics to forecast interaction success metrics and enhance Automatic Speech Recognition (ASR) capabilities
- Promoted to Vertical Lead for the ASK vertical, specializing in third-party skill integration and optimization within the Alexa ecosystem.
- Conducted thorough analysis of various skills using user engagement metrics and performance data, generating detailed reports to inform decisions on optimizing skill enhancements for superior user experience and effectiveness.
- Analyzed user engagement metrics and performance data to identify areas for improvement, resulting in a 20% increase in skill effectiveness and a 15% enhancement in overall user satisfaction through optimized skill enhancements.
- Conducted a detailed comparative analysis of successful and unsuccessful responses within Alexa's shopping vertical, improving overall system performance and enhancing Alexa's responsiveness in the shopping domain.
- Led cross-functional teams in implementing cloud infrastructure improvements for Alexa's shopping vertical, achieving a 30% reduction in response latency and enhancing scalability to meet increasing user demand.
- Engineered a sophisticated error analysis and recovery system for Alexa's Automatic Speech Recognition (ASR) module, leveraging machine learning to identify and correct common misinterpretations, thereby reducing ASR errors by 17% in challenging acoustic environments.
- Implemented a real-time NLG feedback system with sentiment analysis and reinforcement learning, enabling Alexa to adapt responses based on user emotions and interactions. This resulted in a 20% increase in conversation naturalness satisfaction and 15% improvement in perceived empathy.
- Mentored new team members in NLP and Alexa skill development, creating a comprehensive onboarding program that reduced ramp-up time by 40%. This effort improved team productivity and enhanced overall project consistency, fostering a collaborative and efficient work environment.

EDUCATION

The George Washington University

Washington, DC

Master of Science in Data Analytics

Aug 2022- April 2024

Projects:

- **Sign-Language Translator using NN and Deep Learning Algorithms:** Developed a system that translates sign language gestures into text or speech using neural networks and deep learning by using Roboflow tool.
- **Exploratory Analysis on Datasets like Twitter, Yelp and YouTube:** This project demonstrates proficiency in data manipulation, visualization, and deriving actionable intelligence from diverse data sources.
- **Sentimental Analysis on reviews for ChatGPT application:** Implemented a sentiment analysis model to evaluate user reviews of the ChatGPT application, gauging public opinion and user satisfaction.
- **Market simulations and predictions using R and Quarto:** Developed market simulation models and predictive analytics tools using R programming and Quarto for report generation.
- **YouTube Thumbnail Predictor using OpenCV:** Developed an algorithm that analyzes entire video content frame-by-frame using OpenCV to identify the most impactful thumbnail. This tool automatically selects the optimal frame as a thumbnail, enhancing video presentation and potentially increasing viewer engagement and click-through rates for content creators.

Relevant Coursework: Spark Programming, Applied Machine Learning, Data Analysis for Engineers.

Tools used: Python, SQL, R, NLP, Classifiers, Tableau and Power BI.